

German Electricity Demand 2015–2020: Nine Hypotheses Tested Over a 104-Week Forecast

Time Series Modelling Case Study

Link to Repository: [Likhitha25-hub/researchassignment](https://github.com/Likhitha25-hub/researchassignment)

Introduction

This report takes a demand model for electricity in Germany and models it weekly, scoring each model over the same two-year holdout. The data consist of Open Power System Data 60-minute load time series for Germany during the 2015-10 to 2020-09 period, week-mean aggregated time series, where the last 104 weeks (2018-10-05 to 2020-09-25) are the test window and deliberately contain the spring 2020 pandemic demand collapse.

The report is structured into 9 blocks, reflecting the analysis pipeline. As a reader, I find the one-line hypothesis statement at the start of each block useful and the verdict at the end of each block, related to the numbers computed, useful as well. Two of the threads received special attention: the empirical coverage of the 95% prediction intervals – now 72–73% versus the nominal 95% – and the use of the two-stage AIC search to make it possible to use a 147 order SARIMA grid on two CPU cores. The bottom line is bad news for all the trained models: a reproduction of the seasonal naive (the MASE 1.6650) is beat exactly once, by the random forest (at MASE 1.6465), which is a difference of roughly 1%.

Block I Data, Exploration and Stationarity

Hypothesis: *weekly demand is dominated by an annual cycle and needs seasonal treatment before any ARMA structure can help.*

The hourly load was limited to the data from 2015 onwards, converted from UTC to the local time in Berlin, time interpolated for gaps of up to 6 hours, and scaled from MW to GW. This left 50400 hourly observations, 2101 daily means and 299 weekly means. The aggregation was done once a week, with a Friday anchor, and boundary weeks that contained less than 160 hours were discarded to prevent a half full week affecting the weekly statistics. The weekly series is of average value of 55.5 GW and a range of 63.3 to 46.0 GW.

Three time scales are shown in figure 1. The weekly panel shows winter peaks of up to about 63 GW, summer lows at the end of the low fifties and a strong dip every Christmas week, and a fall in spring 2020, that has no precedent in earlier years. The weekday–weekend cycle is included in the daily panel and the hourly panel (January 2018) illustrates the double-peaked working day. There are at least three seasonal layers in the demand therefore: the annual one (period 52) is the modelling target at weekly resolution.

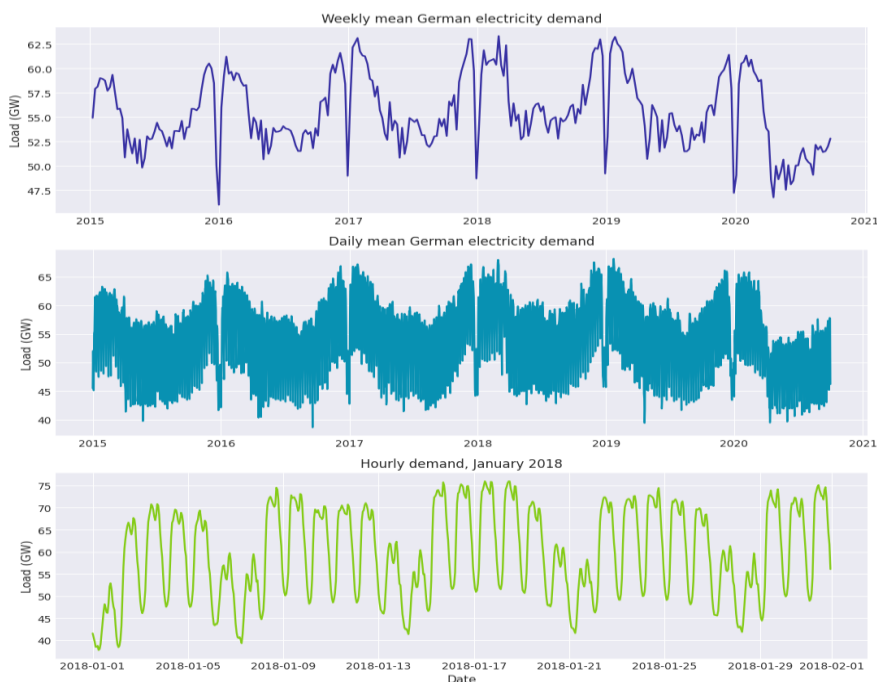


Figure 1. Weekly, daily and hourly views of German electricity demand. The weekly panel shows the annual cycle, festive dips and the spring 2020 drop.

The level series strongly rejects a unit root (ADF statistic -3.87 , $p = 0.0023$); the seasonal difference does not reject a unit root ($p = 0.124$), but the first difference and first + seasonal difference both reject strongly ($p < 0.001$). The level test isn't enough, as the ACF in Figure 2 reveals that the correlations slowly fade and then rise back to a peak at lag 52. One regular difference and a seasonal difference almost exactly cancel; the majority of the correlations are within the confidence band, with a negative spike at lag 52 the characteristic "kick" of a seasonal moving-average term.



Figure 2. ACF and PACF of the level series (top) and after first plus seasonal differencing (bottom). The lag-52 peak motivates $D = 1$, $s = 52$.

Verdict confirmed the annual cycle dominates, and $d = 1$, $D = 1$, $s = 52$ became the working differencing scheme.

Block II Evaluation Design

Hypothesis: one fixed test window and one common error scale make nine very different models comparable.

The first 195 weeks were used for training and the last 104 weeks were held out. Four statistics are calculated for each forecast, including MAE, RMSE, MASE and mean bias (forecast - actual). The ratio of test MAE to in-sample MAE of seasonal-naïve forecast of training years is 1.385 GW for test here, and a self-check showed a perfect forecast scores exactly zero. A MASE below 1 indicates that the model is more accurate than the seasonal-naïve method, also calculated using the training data (Hyndman & Athanasopoulos, 2021). Bias remains as the test window is the one where the structural fall in demand is included, and the sign of the average error indicates whether a model over- or underestimated the pandemic period. For interval forecasts, we also provide empirical coverage, that is, the percentage of the test actuals that are contained in the 95% interval. Calibrated intervals should be approximately 95% and the lack of intervals has significance in operations as the reserve sizes are based on these bands (Hong & Fan, 2016).

Verdict design fixed every number that follows is computed on the same 104 weeks against the same 1.385 GW scale.

Block III Benchmark Forecasts

Hypothesis: a copy of last year is the strongest cheap forecast for a strongly seasonal series.

The four benchmarks that were run were: mean, naïve, seasonal naïve and drift. The final training year has been repeated in the seasonal naïve model, which scored MAE 2.3060 GW, RMSE 3.0450, MASE 1.6650 and bias +1.7368. The others were far behind: naïve MAE 3.8906 (MASE 2.8092), drift 4.0593 (2.9310), mean 3.7855 (2.7332). Only the seasonal naïve lands in the winter peak and the festive trough on the right weeks; the flat and trend benchmarks slice through the seasonal swing, as shown in Figure 3. That its MASE is 1.67 and not 1.0 indicates how difficult the test window has been compared to the training years – that is, it has a positive demand drop that no repetition of 2017–18 can foresee.

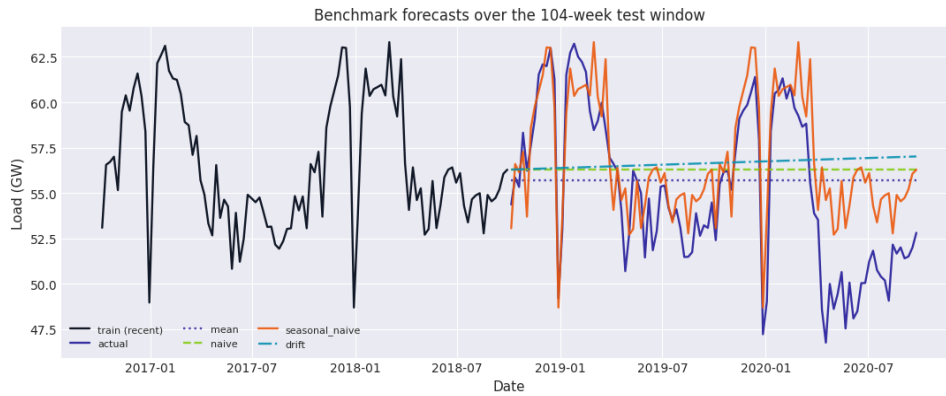


Figure 3. Benchmark forecasts over the 104-week test window. The seasonal naive is the only benchmark that reproduces the seasonal shape.

Verdict confirmed the seasonal naive sets the bar at 2.3060 GW MAE, and every model below is judged against it.

Block IV SARIMA: Two-Stage Grid Search and Diagnostics

Hypothesis: a tuned SARIMA should beat a copy of last year.

It was impossible to fit 147 seasonal state-space models on the equipment used in the search, namely two Colab cores, so the search was executed in two stages. Stage 1 was an all order stage with the optimiser limited to 120 iterations, and stage 2 was performed with the full default settings, only refitting the winning order from stage 1. The cap is for ranking, not reporting – all 147 candidates were given an equal iteration budget, with the partially converged likelihoods thus being compared on an equal basis, and the AIC ordering converging well before that point. The result is that the two AIC values cannot be mixed up between stages; the winning order (6,1,6) screened at 298.925 and refit at 311.057, not because there was any change in the model, but because of the different budgets for the optimiser. We quote 311.057 as the final AIC. The total search time was 109.9 minutes with the runner-up (1,1,6) at 300.865 so that the moving-average structure with $q = 6$ is more than high AR order and carries the fit.

The result is moderated by diagnostics. Even the best in the family leaves correlation on the table in the residual ACF (Figure 4); the residual histogram has a long tail to the left from festive weeks during which the model overshoots.

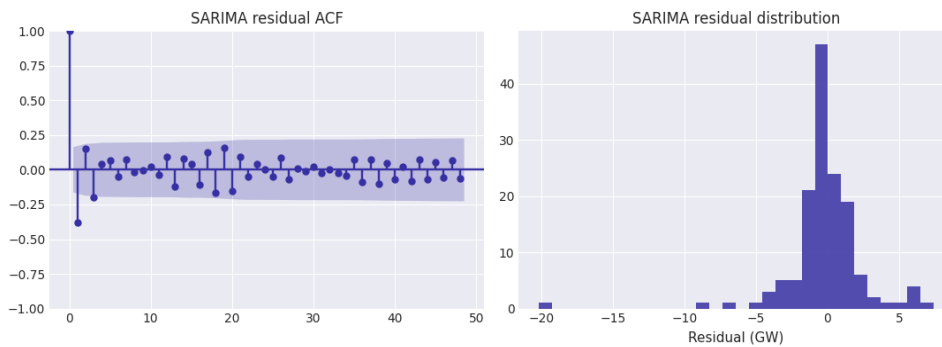


Figure 4. SARIMA (6,1,6)×(1,1,1,52) residual ACF and distribution. Ljung–Box rejects whiteness at lag 24 ($p = 0.0001$).

On the test window (Figure 5) the model scored MAE 3.2219, RMSE 3.9879, MASE 2.3263 and bias +3.1028. It continues the seasonal pattern of 2018–19 and moves above the actuals in 2020, as it is simply an extension of the pre-pandemic dynamics. Only 72.1% of test weeks were within the 95% interval. In both failures, the parameters and interval widths were learned from the years that had no structural change, and the bands are narrowest at exactly the places where the mean is most incorrect.

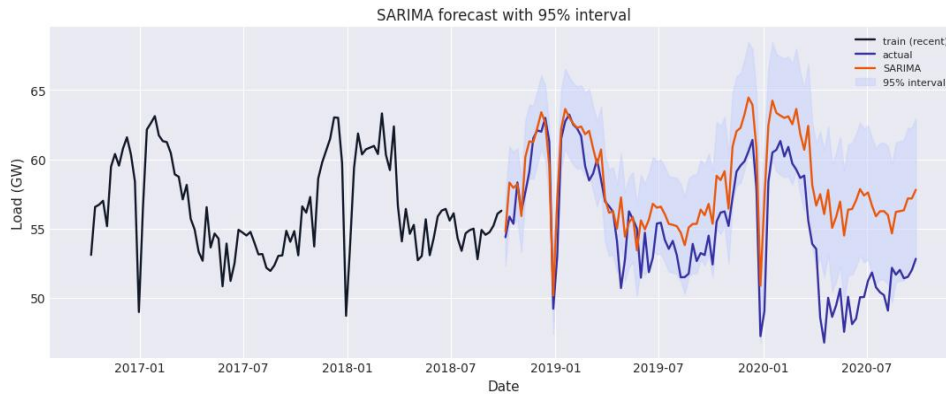


Figure 5. SARIMA forecast with 95% interval. Empirical coverage is 0.721 against the nominal 0.95.

Verdict *rejected the tuned SARIMA (MASE 2.3263) loses to the seasonal naive (1.6650), and its intervals under-cover by 23 percentage points.*

Block V SARIMAX: A Conditional Temperature Forecast

Hypothesis: *adding temperature information should sharpen the SARIMA forecast.*

Heating and cooling degree-days (base 15.0 °C and 22.5 °C) were calculated from weekly mean daily Berlin temperature from the Open-Meteo archive. Each exogenous column was time-shifted one week prior to being used in the model as week t is always being predicted based on weather that occurred in week $t-1$ the leakage guard. Since observed, rather than forecast temperatures are provided across the test window, the output is a conditional forecast, not an operational forecast. With reuse of the order (6, 1, 6), SARIMAX had worse MAE of 3.4442, MASE of 2.4868 and bias of +3.3423 (Figure 6) on all the point metrics as compared to plain SARIMA; however, the coverage did improve marginally to 73.1%. This degradation has been explained: the weekly averaging smooths most of the temperature signal not already removed by seasonal differencing, while the lagged weather provides more parameters and noise than information.

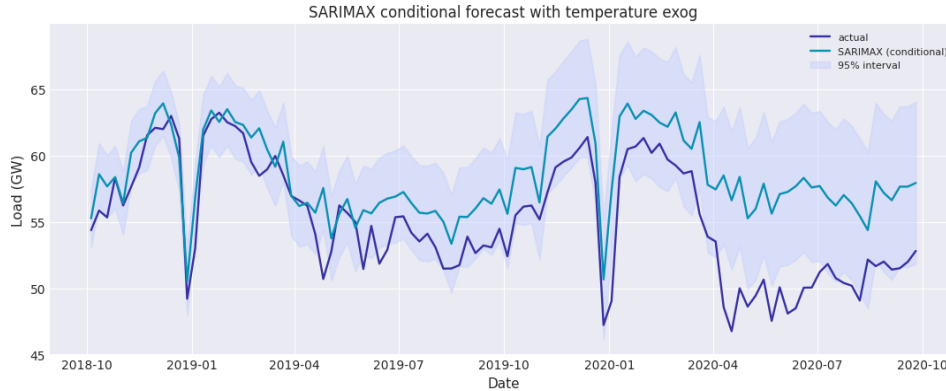


Figure 6. SARIMAX conditional forecast with lagged temperature, HDD and CDD regressors. Coverage 0.731.

Verdict *rejected conditioning on last week's weather cost 0.22 GW of MAE and bought a single point of coverage.*

Block VI Random Forest with a Recursive Rollout

Hypothesis: *a feature-based model that sees lag 52 directly should at least match the seasonal naive.*

The supervised table contains 13 features: week-of-year sine and cosine, month, number of holidays, the shifted temperature three, target lags 1, 4, 13 and 52, and moving averages over 8 and 26 weeks, after warm-up losses. A random forest with 400 trees and a histogram gradient-boosting model with 400 iterations were fit; forecasting is recursive, meaning that the forecasted week is added to the history and future lag and running features are created from the forecasts, not from the actuals. The forest scored MAE 2.2803, RMSE 2.9699, MASE 1.6465 and bias +1.9132 the only model in the study to beat the seasonal naive, at a MASE ratio of 0.9889. We read that 1% margin as "marginal" and not "significant": it is based on just one test window and no significance test, and could easily be just the opposite with a different split. The boosting model reached MAE 2.6058 (MASE 1.8815).

The near tie is explained in figure 7. About 0.8 of the total importance is from the lag_52 feature, the forest is basically a seasonal naive with small adjustments from holiday counts and most recent lags, and that adjustment is the 1%. Both of the tree models flatten above the actuals in spring 2020 as all of the models did (see Figure 8).

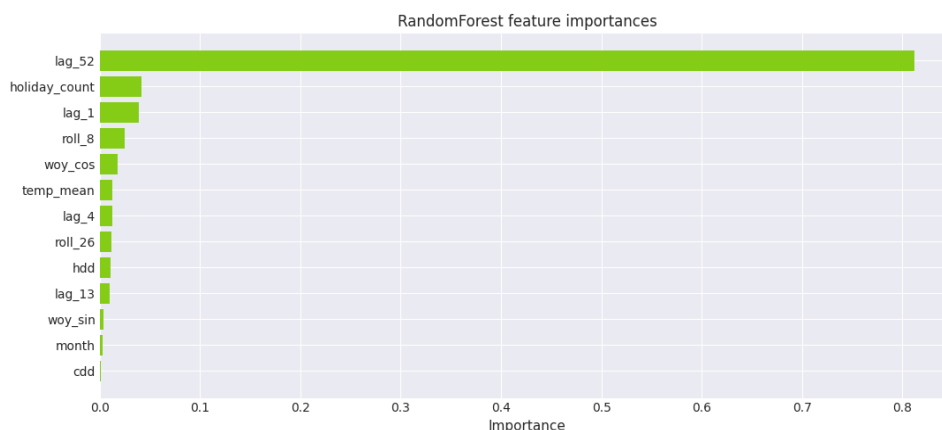


Figure 7. Random forest feature importances. lag_52 dominates, with holiday_count and lag_1 next.

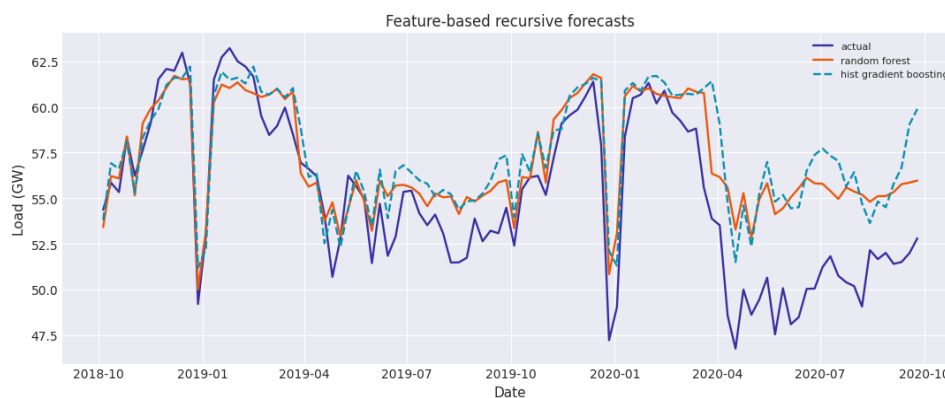


Figure 8. Recursive forecasts of the two tree models against the actual test series.

Verdict confirmed on paper, marginal in substance MASE 1.6465 against 1.6650.

Block VII LSTM on Hourly Data

Hypothesis: an hourly sequence model can convert fine-grained structure into a better weekly forecast.

The hourly series was divided at the same date and only standardised using training statistics. With a 168-hour look-back the training set framed into 27,846 windows plus 4,914 for validation. A sweep over units {48, 96} and dropout {0.0, 0.2} at learning rate $5e-4$ with early stopping selected (48, 0.0, 0.0005) at validation loss 0.01186. The forecast is a true free run: 17,472 steps per hour with each prediction fed into the window and executed by a compiled tf.function step at ~ 7.5 ms per step. When aggregated to weekly, it scored MAE 3.7926, RMSE 4.4029 and MASE 2.7384, which were worse than all models except naive and drift.

As can be seen in figure 9, the reason is that the rolloff approaches a near level curve around 55.8 GW. One-step training provides no opportunity to keep it sharp over seventeen thousand self-fed steps a known training failure mode of recursive neural forecasting (Hewamalage et al., 2021). The smallest bias (+0.5811) is among the worst, because the line of its bias is flat near the middle of the test distribution, thus balancing over- and under-shooting without following anything.

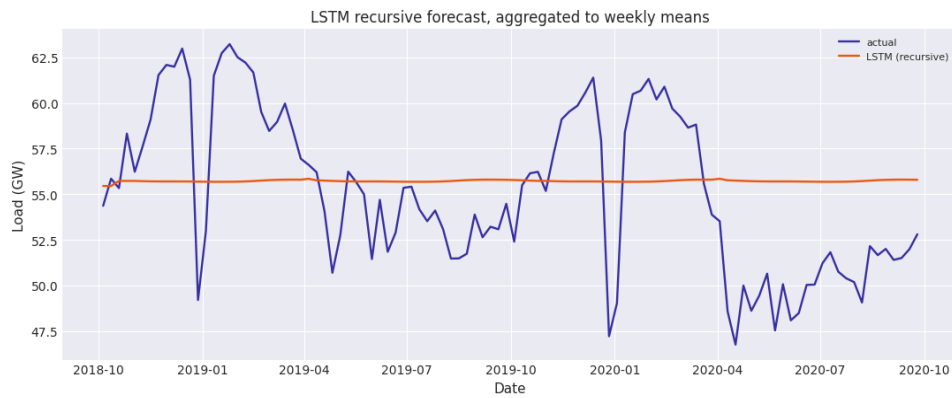


Figure 9. LSTM free-run forecast aggregated to weekly means. The recursive rollout collapses toward a constant level.

Verdict *rejected MASE 2.7384; the near-zero bias is an artefact of flatness, not calibration.*

Block VIII Consolidated Results

Hypothesis: *assembled in one table, the ranking is decided by how each model handles the seasonal shape and the 2020 break.*

Table 1. Test-window metrics for all models (GW where applicable), sorted by MASE. Ratio is MASE relative to the seasonal naive.

Model	MAE	RMSE	MASE	Bias	Ratio
random_forest	2.2803	2.9699	1.6465	+1.9132	0.9889
seasonal_naive	2.3060	3.0450	1.6650	+1.7368	1.0000
hist_gbr	2.6058	3.3567	1.8815	+2.2874	1.1300
sarima	3.2219	3.9879	2.3263	+3.1028	1.3972
sarimax	3.4442	4.2528	2.4868	+3.3423	1.4936
mean	3.7855	4.3960	2.7332	+0.5555	1.6416
lstm	3.7926	4.4029	2.7384	+0.5811	1.6447
naive	3.8906	4.5074	2.8092	+1.1402	1.6872
drift	4.0593	4.7229	2.9310	+1.5092	1.7603

Two models sit under MASE 1.7 and both are essentially seasonal repetition one literal, one learned. Every trained model carries a positive bias, from +1.91 (forest) to +3.34 (SARIMAX), because all of them project pre-2020 demand into a pandemic year; the two smallest biases (mean, LSTM) belong to flat forecasts that split the difference rather than track the series. Figure 10 overlays all nine forecasts.

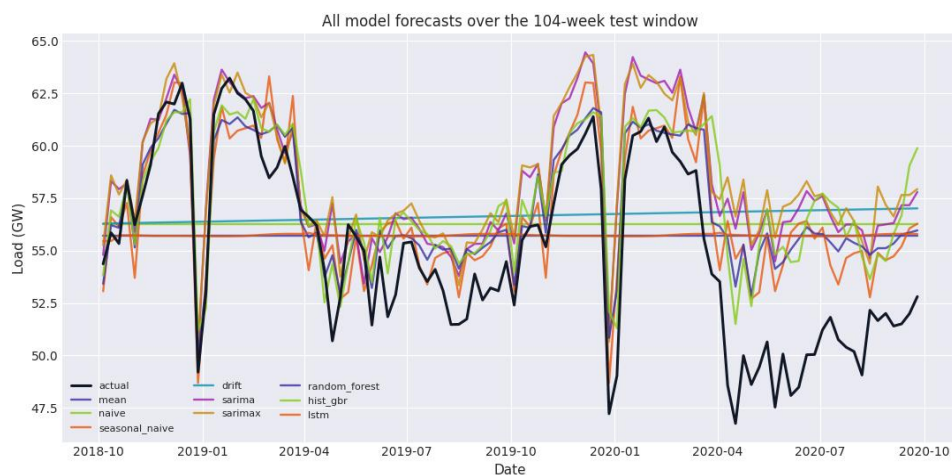


Figure 10. All model forecasts over the 104-week test window against the actual series (black).

Verdict *supported every fitted model is behind at least one untrained benchmark; only the forest edges the seasonal naive.*

Block IX Regime Deep Dive

Hypothesis: averages hide where models fail; festive and temperature-extreme weeks should separate them.

The MAE was re-computed on three regimes: festive ISO weeks (52, 1, 2), the five coldest test weeks and the five hottest test weeks based on mean temp (Table 2). The seasonal naive wins the festive weeks (1.7520) and the hottest weeks (1.2450) where the weeks always fall at the same time of year, which is what a copy of last year does. Seasonal naive (2.3041) is worst, followed by the boosting model (1.3440) and the forest (1.7469), because the cold snaps do not occur on the same ISO week every year, and the tree models can respond to these through temperature and lag features. Where the LSTM is most needed is at “festive” (5.4713) and at “coldest” (6.5962), where a flat trajectory misses every excursion.

Table 2. Regime MAE (GW): festive ISO weeks 52/1/2, five coldest and five hottest test weeks.

Model	Festive	Coldest 5	Hottest 5
seasonal_naive	1.7520	2.3041	1.2450
random_forest	2.0346	1.7469	1.8931
hist_gbr	2.2129	1.3440	1.7017
sarima	3.5838	2.0544	1.9701
sarimax	3.6530	2.5052	2.5117
lstm	5.4713	6.5962	2.4399
naive	5.6671	6.2435	2.9828
drift	5.7562	6.2576	3.4383

Verdict confirmed no model wins all three regimes; naive-versus-forest is a choice between calendar-locked and weather-driven errors.

Discussion: The Six Case-Study Questions

Q1 Improvement over the seasonal naive indicator. None that we would consider having any significance. The only model below the benchmark (MASE 1.6465 vs 1.6650, ratio 0.9889) is the random forest and a 1% margin on a single 104-week window (without a significance test) is a plausible split-to-split noise. All other models perform 13–76% worse, with the tuned SARIMA performing 40% worse. On this series, there is no need to be sophisticated, calendar memory is sufficient.

Q2 Avoiding leakage in temperature features. Weather up to week $t-1$ was used for a forecast for week t , and temperature, HDD and CDD were only introduced into models after a 1 week shift. The features of the tree-model lag and rolling were constructed using shifts, and the recursive rollout is based on the model's own forecasts, and not actual test data. Only training statistics were used for the LSTM scaler. There is no need to make any adjustments to the number of holidays as the calendar is known ahead of time.

Q3 Differencing orders and seasonal period. $s = 52$ because weekly resolution gives the annual cycle at lag 52 which is where the ACF peaks. The addition of $d = 1$ makes this peak stationary at the level ACF, whereas the seasonal difference alone does not ($p = 0.124$); the addition of $d = 1$ again gives good stationarity ($p < 0.001$), and the screening table indicates that the best candidates all involve $d = 1$. The combined (1,1,52) scheme also outputs a lag-52 MA signature which is incorporated into the seasonal MA term.

Q4 - Are holiday and temperature covariates helpful? However, the linear model, SARIMAX, did not benefit from the temperature, with a gain of one coverage point (0.731 vs 0.721) and an accuracy loss (MASE 2.4868 vs 2.3263). Where a model could use the information on a nonlinear basis, holiday information proved useful holiday count is the second most important forest feature, and the forest is the one model that outperformed the benchmark. Availability: holidays are known at the forecast origin, observed temperature is not; hence the SARIMAX run is conditional. An operational version would take over the weather-forecast error, in addition to what we measured.

Q5 Interpretability and complexity. SARIMAX is most readable: a dozen coefficients with signs and standard errors, but a 109.9-minute grid search and still failed its own residual test. The forest is dense tree by tree: it is honest in its global explanation via importances (lag_52 at ~ 0.8), and it fits in minutes. The LSTM model was the least interpretable and most costly to tune model, and provided the lowest usable forecast. Complexity and accuracy were inversely related with 195 weekly training points.

Q6 Operational use recommended. The seasonal naive. Reliability: within 1% of the best model on MASE. Uncertainty: it ships without model intervals, fitted alternatives are shipping intervals with miscalibrated intervals (72-73% real coverage on a 95% label), and an empirical or conformal band around the seasonal naive would be more credible than either. Interpretability: complete. Zero maintenance (no refits, no

feature-pipeline, no weather-feed). Once the margin of the forest is made reliable, the forest becomes the recommendation; improvements below point to this.

The pandemic by the numbers. Biases range from +1.74 (seasonal naive) to +3.34 (SARIMAX) for all models as the test window includes a structural break does not present in all the training years. The +0.58 on the LSTM is ideal at present, but it is not a prediction of the future, rather a lack of direction. The measured bias is partly due to the transient cost of an unforecastable shock since, by the end of the window, the actual series returns to the forecasts.

Future Improvements

Three flaws in this run point clearly signal fixes. First, the first interval under-coverage (0.721 and 0.731 against the nominal coverage of 0.95) is the strongest argument for distribution-free calibration: conformalized quantile regression wraps any point model including the forest, which currently has no intervals, in bands with finite-sample coverage guarantees (Romano, Patterson and Candes, 2019), and interval quality should then be scored with proper scoring rules such as the interval score (Gneiting and Raftery, 2007) rather than coverage alone. Second, Ljung–Box rejection at (6,1,6) ($p = 0.0001$) indicates that seasonal differencing at $s = 52$ is a crude device for this series; dynamic harmonic regression, Fourier terms for the annual cycle with ARMA errors, models the same seasonality with much fewer implicit parameters and (as described earlier) adds break dummies for 2020 (Hyndman and Athanasopoulos, 2021). Third, the forest's 1% margin should be either consolidated or dismissed: Averaging the forest with the seasonal naive is a natural variance-reduction step in the spirit of bagging (Breiman, 2001), and the comparison should be repeated using multiple rolling forecast origins to provide the forest margin with a distribution rather than a point value. The free-run collapse suggests that 17,472 recursive steps may not be the best approach for the LSTM, however, and that direct multi-horizon training or sequence-to-sequence output with the calendar covariates as input would be better options (Hewamalage, Bergmeir and Bandara, 2021). Lastly, the entire assessment needs to be shifted from point to probabilistic forecasting where the load-forecasting practice has converged (Hong and Fan, 2016).

Limitations

The conclusions are based on a single 104-week test window which encompasses a pandemic; for instance, the 1% margin in the forest may not withstand other windows. The screening step assumes that the ranking obtained by the AIC iteration with 120 iterations is equal to the fully converged ranking (which is only indirectly verified through the winner's refit). The SARIMAX result is conditioned on the observed temperature, and for the tree models, which use the realised one-week-lagged temperature over the test window, so that is a bit of a coincidence the size of the margin above the seasonal naive it could just be a matter of conditioning. The within-week data the LSTM was intended to leverage is lost every week by aggregation, and its recursive nature exacerbates this issue. The metric differences were not subjected to any statistical test; hence the forest's victory is not real but "marginal."

Conclusion

In nine hypotheses, the survivals are those which are modest in scope. The seasonal naive set a bar (MASE 1.6650) that a 147-order SARIMA search (2.3263), a temperature-conditioned SARIMAX (2.4868) and an hourly LSTM (2.7384) all failed to beat, and that a random forest did by about 1% (1.6465), mostly by rediscovering lag 52. The most significant discovery isn't the ranking, but the calibration - both interval-producing models had 72–73% coverage of test weeks using a 95% label. The next step in this series isn't a larger version; it's reliable uncertainty in a simple version.

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